

No notes, calculators, or other aids are allowed. Read all directions carefully and write your answers in the space provided.

Taking this exam signifies you are aware of and in accordance with the Academic Honor Code of Georgia.

Do not separate any pages from the rest of your exam.

Version Ryland Grace
100 points

For the following questions, please completely fill in the bubble corresponding to your chosen answer.

- [3] 1. (new) Because a shift cipher has many possible keys, trying every possible key is not a practical method of breaking the cipher.

☐ True

☐ False

- [3] 2. (new) For every integers $n \geq 0$ and $0 \leq r \leq n$, it holds

$$C(n, r) \leq 2^n$$

☐ True

☐ False

- [3] 3. (new) The number of length-eight binary strings containing exactly two 1s is equal to the number of three-element subsets of an eight-element set.

☐ True

☐ False

- [3] 4. (new) A student must choose exactly one elective from three categories. The first category contains n_1 courses, the second contains n_2 courses, and the third contains n_3 courses. The total number of possible choices is $n_1 + n_2 + n_3$.

☐ True

☐ False

- [3] 5. (new) For $0 \leq r \leq n$, the number of length- n binary strings containing exactly r ones equals the number containing exactly r zeros.

☐ True

☐ False

- [5] 6. (new) Which of the following statements about RSA is correct?
- ☐ The security of RSA relies on keeping the modulus n secret.
 - ☐ Anyone who knows the public key can efficiently compute the private key.
 - ☐ RSA is secure because multiplying large prime numbers is computationally difficult.
 - ☐ Factoring the modulus n would allow an attacker to compute the private key.
 - ☐ None of the above.
- [5] 7. (new) The *Hail Mary* records 37 diagnostic measurements, each assigned to exactly one of 8 categories. What is the largest number of measurements guaranteed to belong to the same category?
- ☐ 5
 - ☐ 4
 - ☐ 6
 - ☐ 8
 - ☐ None of the above.
- [5] 8. (new) Grace distributes 18 identical emergency food packets among four distinct ship modules. The command module must receive at least 4 packets, and each other module must receive at least 1 packet. How many distributions are possible?
- ☐ $C(21, 3)$
 - ☐ $C(17, 3)$
 - ☐ 4^{18}
 - ☐ None of the above.
 - ☐ $C(14, 3)$
- [5] 9. (new) A *Hail Mary* diagnostic procedure has four stages. The stages can be completed in 3, 2, 4, and 5 ways, respectively, and the choice at one stage does not restrict the choices at the others. How many complete procedures are possible?
- ☐ 14
 - ☐ 120
 - ☐ 60
 - ☐ 625
 - ☐ None of the above.
- [5] 10. (new) Grace selects 5 reports for a mission archive from 4 astrophysics reports, 5 engineering reports, and 3 medical reports. The archive must contain exactly 2 astrophysics reports and exactly 1 medical report. How many selections are possible?
- ☐ 60
 - ☐ 120
 - ☐ 180
 - ☐ 360
 - ☐ None of the above.

- [5] 11. (new) Nine distinct sample canisters are arranged in a row aboard the *Hail Mary*. Three specified canisters must appear consecutively, although they may appear in any order within their block. How many arrangements are possible?
- ☐ 7!3!
 - ☐ 7!
 - ☐ $9!/3!$
 - ☐ $9! - 3!$
 - ☐ None of the above.
- [5] 12. (new) How many distinct arrangements of the word SCIENCE have all three vowels consecutive? The consonants are not required to be consecutive.
- ☐ 60
 - ☐ 120
 - ☐ 180
 - ☐ 360
 - ☐ None of the above.
- [5] 13. (new) Grace, Rocky, and four other distinct crew members sit at a circular table with six evenly spaced seats. In how many rotationally distinct seatings do Grace and Rocky sit directly opposite each other?
- ☐ 12
 - ☐ 48
 - ☐ 120
 - ☐ None of the above.
 - ☐ 24
- [5] 14. (new) Grace distributes 12 identical backup fuel cells among three distinct ship modules. Each module must receive at least 2 cells and at most 6 cells. How many distributions are possible?
- ☐ 16
 - ☐ 19
 - ☐ 21
 - ☐ 28
 - ☐ None of the above.

Instructions: For the following question(s), you must show **all** of your work using methods covered in lecture. Brute-force solutions will receive **no credit**. The only exception is that you may use the brute-force method to find multiplicative inverses. All other parts of the problem must be solved using techniques taught in class.

- [13] 15. Solve the following system of congruences. Give the smallest nonnegative solution.

$$x \equiv 3 \pmod{7}$$

$$x \equiv 8 \pmod{11}$$

Show all work using the Chinese Remainder Theorem.

- [13] 16. (new) During a communications test, the *Hail Mary* sends 101 transmissions through 20 available channels. Prove that at least one channel carries at least 6 transmissions.

[14] 17. The *Hail Mary* carries 16 distinct scientific samples. Give a combinatorial proof of the identity

$$\sum_{k=4}^{12} \binom{16}{k} = 2^{16} - 2 \sum_{k=0}^3 \binom{16}{k}.$$

In your proof:

1. Explain what the left-hand side counts and how it counts those objects.
2. Explain what the right-hand side counts and how it counts those objects.
3. Explain why the two sides count the same collection of objects.

18. **EXTRA CREDIT:** Design a mission patch for the *Hail Mary* that includes both Earth and Rocky's home star, 40 Eridani. Submissions will be judged on a scale of 1–3. You may not raise your hand to ask a TA anything concerning this question.

Scratch Paper: If using this page to write a solution, clearly indicate on both this page and the original page that the solution is here.